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# DSC5001: Statistical Machine Learning I

## Tutorial 5: Polynomial and Spline Regression

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# Tutorial Setup

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Complete Tasks 1–3.

Task 4 is optional.

- ▶ **Task 1:** Bias–variance and prediction error
- ▶ **Task 2:** Cubic spline and B-spline calculations
- ▶ **Task 3:** Cross-validation and model selection
- ▶ **Task 4:** Python comparison of polynomial and spline models

For Tasks 1–3, show the main calculations and explain the statistical meaning of the result.

# Task 1: Bias–Variance and Prediction Error

At a fixed input  $x_0$ , suppose

$$f(x_0) = 5, \quad \sigma^2 = 1.$$

Two learning procedures have the following repeated-sample properties:

|         | $\mathbb{E}_D[\hat{f}_D(x_0)]$ | $\text{Var}_D[\hat{f}_D(x_0)]$ |
|---------|--------------------------------|--------------------------------|
| Model A | 4.5                            | 0.5                            |
| Model B | 5.2                            | 1.4                            |

Answer the following:

1. Compute the bias and squared bias of each model.
2. Compute the expected squared prediction error of each model.
3. Which model has the smaller expected prediction error?
4. If Model B has lower training error, does that change your answer to part 3? Explain.

## Task 2: Cubic Spline and B-Spline Basis

Consider cubic splines on  $[0, 2]$  with one interior knot  $\xi = 1$ . Here  $K = 1$ , so  $M = K + 4 = 5$ . The truncated-power basis is

$$\phi_T(x) = [1, x, x^2, x^3, (x - 1)_+^3]^\top.$$

A B-spline basis for the same spline space is obtained by

$$B(x) = C\phi_T(x), \quad C = \begin{bmatrix} 1 & -3 & 3 & -1 & 1 \\ 0 & 3 & -9/2 & 7/4 & -2 \\ 0 & 0 & 3/2 & -1 & 2 \\ 0 & 0 & 0 & 1/4 & -2 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

- (a) Compute  $B(0.5)$  and  $B(1.5)$ . Which basis functions are zero at each input?
- (b) Explain briefly how the results in (a) illustrate **local support**.
- (c) Let

$$\theta = [1, 2, 4, 6, 3]^\top, \quad f(x) = \theta^\top B(x).$$

Find the equivalent truncated-power coefficient vector  $\beta$  such that

$$f(x) = \beta^\top \phi_T(x).$$

- (d) Verify at  $x = 0.5$  that the two representations give the same prediction.

## Task 3: Cross-Validation and Model Selection

**4-fold cross-validation (CV):** fit on three folds and evaluate on the remaining fold, rotating the held-out fold. Each validation fold contains 10 observations. The table gives training MSE and validation SSE.

| Model           | Train MSE | $F1$ | $F2$ | $F3$ | $F4$ |
|-----------------|-----------|------|------|------|------|
| Degree 2        | 2.10      | 24   | 20   | 22   | 26   |
| Degree 5        | 1.30      | 18   | 25   | 21   | 28   |
| Degree 9        | 0.45      | 11   | 30   | 27   | 32   |
| Spline, 3 knots | 1.15      | 17   | 19   | 18   | 20   |

For each fold,

$$\text{MSE}_{\text{fold}} = \frac{\text{SSE}_{\text{fold}}}{10}.$$

- (a) Convert the four SSE values into fold MSEs and compute the mean CV MSE for each model.
- (b) Compute

$$\text{CV gap} = \text{CV MSE} - \text{training MSE}.$$

Which model shows the strongest sign of overfitting?

- (c) Which model should be selected by CV?
- (d) Degree 2 and degree 5 have the same CV MSE. If these were the only two candidates, which would you choose, and why?
- (e) After model selection, how should the final test set be used?

## Optional Task 4: Polynomial versus B-Spline in Python

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Generate a nonlinear regression dataset using

$$Y = 2 + \sin(3X) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 0.3^2).$$

Compare:

- ▶ degree-3 polynomial regression;
- ▶ degree-7 polynomial regression;
- ▶ cubic B-spline regression with 6 knots.

Use the same 5-fold cross-validation split for all three models.

1. Report the mean cross-validation MSE.
2. Select the model with the smallest CV MSE.
3. Fit all three models to the full dataset.
4. Plot the fitted curves together with the observations.

Use a fixed random seed so the result is reproducible.

# Tutorial Summary

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After this tutorial, you should be able to:

- ▶ use the bias–variance decomposition to compare prediction error;
- ▶ evaluate truncated-power spline features;
- ▶ compute predictions from supplied B-spline basis values;
- ▶ use cross-validation to choose model complexity;
- ▶ compare polynomial and B-spline regression computationally.

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## **Tutorial 5: nonlinear regression, splines, and model selection**

Questions and discussion